



DNA Sequence Analysis Tool using Boyer–Moore String Matching Algorithm

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ABSTRACT

GeneScan is a DNA Sequence Analysis Tool developed using the Boyer–Moore String Matching Algorithm to efficiently search for specific DNA patterns or gene motifs. The system allows users to upload DNA sequences and search for desired patterns within them. It identifies and highlights matching regions while displaying the search time and number of character comparisons. GeneScan shows how advanced DSA string-matching techniques can be applied to real-world areas such as bioinformatics, genetics, and medical research.

PROBLEM STATEMENT

DNA sequences are very large, making it difficult and time-consuming to search for specific gene patterns using traditional methods. This project aims to develop a **DNA Sequence Analysis Tool** using the **Boyer-Moore String Matching Algorithm** to efficiently search for DNA motifs and display the results with improved **speed and accuracy**.

PROPOSED FUNCTIONALITIES



1. **DNA Sequence Input** – Enter a DNA sequence using A, T, G, and C.
2. **Pattern Input** – Enter the DNA pattern to be searched.
3. **Pattern Searching** – Use the **Boyer–Moore algorithm** to search the pattern efficiently.
4. **Find All Matches** – Display all positions where the pattern occurs.
5. **Sequence Validation** – Check whether the DNA sequence contains only valid bases.
6. **Result Display** – Show whether the pattern was found and display the matching

TOOLS AND TECHNOLOGIES TO BE USED

1. **Programming Language :Java** – Implementation of Boyer–Moore **Core Algorithm** : Boyer–Moore String Matching
2. **File Handling : FASTA / TXT** – DNA sequence input
3. **Data Structures : Arrays / Strings** – DNA sequences and pattern processing
4. **Frontend / User Interface** : HTML, CSS, JavaScript – User interface and result visualization
5. **Development Tools** : VS Code / IntelliJ IDEA – Code development
 - a. **Git & GitHub** – Version control and project management



🎯 **Goal:** Fast and efficient search of DNA patterns using Boyer-Moore String Matching Algorithm. 🌟 **Skills Demonstrated:** Advanced String Algorithms, File Handling, Efficient Searching, Data Visualization

OUTCOME

- ~ Quickly finds specific DNA and gene patterns using the Boyer–Moore algorithm.
Identifies matching positions and helps detect important DNA patterns such as mutations and disease markers.
- ~ Compares Boyer–Moore with Naïve String Matching to show search efficiency.
- ~ Can be used in genome comparison, CRISPR sequence searching, and genetic research.
- ~ Demonstrates how DSA and string-matching techniques can be applied to real-world bioinformatics problems.



Thank You